



SQL Project Music Store Analysis

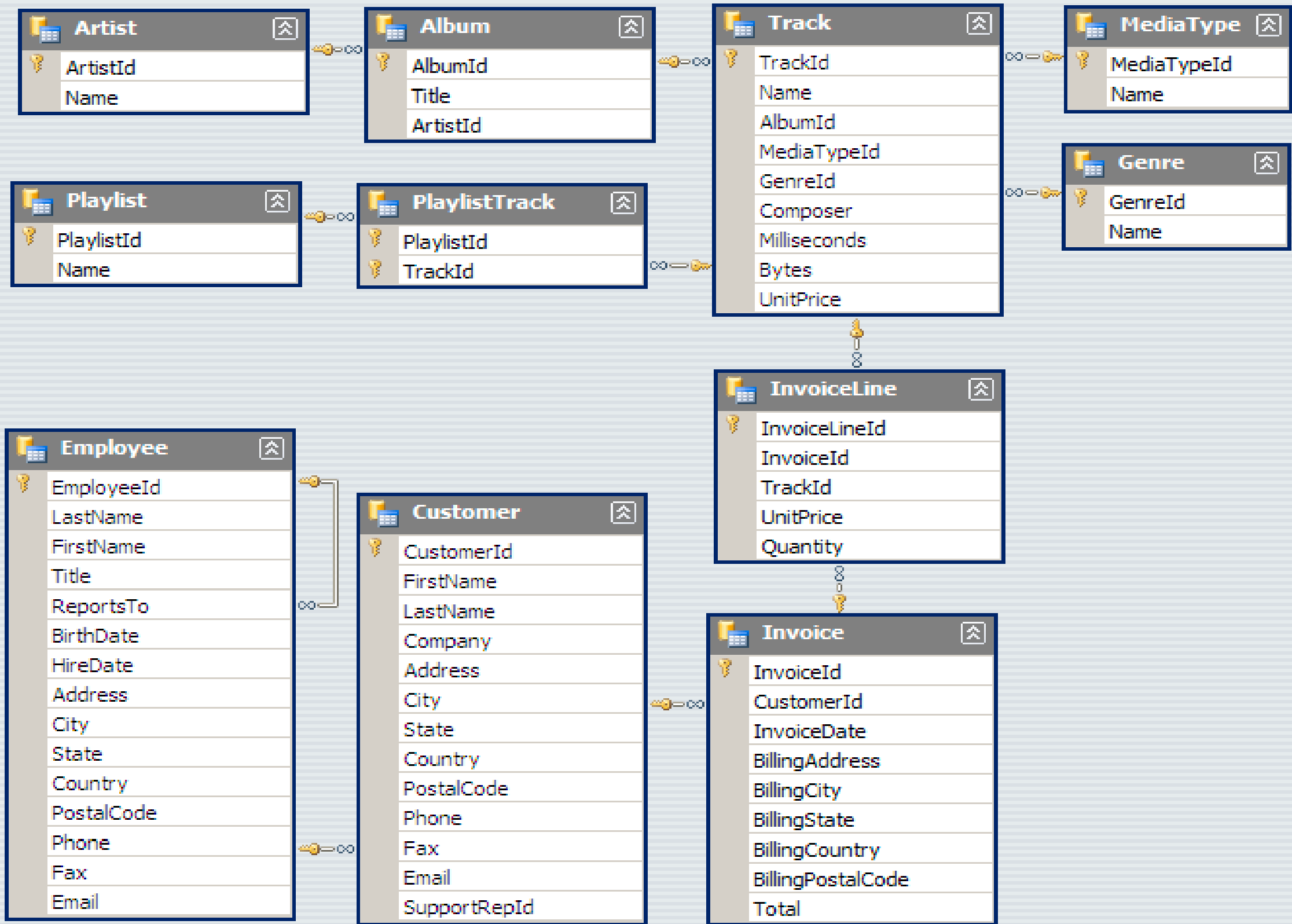
RDBMS used to run the queries : **PostgreSQL**

OBJECTIVE :

The primary objective of the music store is to achieve sustainable business growth while addressing existing challenges.

We need to examine the dataset with SQL and help the music store understand its business growth by answering simple questions.





MUSIC PLAYLIST DATABASE SCHEMA



DIVISION OF QUESTIONS

Questions are divided into 3 categories based on the complexity of the questions

EASY

MODERATE

ADVANCE

Question 1 (EASY)

Who is the senior-most employee based on job title?

```
---Q1: Who is the senior most employee based on job tilte?  
select last_name,first_name,title,levels  
from employee  
order by levels desc  
limit 1
```

Output:

	last_name character (50) 	first_name character (50) 	title character varying (50) 	levels character varying (10) 
1	Madan ...	Mohan ...	Senior General Manager	L7



Question 2 (Easy)

Which country has the most invoices?

```
---Q2:Which countries have the most invoices?  
select count (*) as c, billing_country  
from invoice  
group by billing_country  
order by c desc
```



Output:

	c bigint 	billing_country character varying (30) 
1	131	USA
2	76	Canada
3	61	Brazil
4	50	France
5	41	Germany
6	30	Czech Republic
7	29	Portugal
8	28	United Kingdom
9	21	India
10	13	Chile
11	13	Ireland
12	11	Spain
13	11	Finland
14	10	Australia
15	10	Netherlands
16	10	Sweden
17	10	Poland
18	10	Hungary
19	10	Denmark
20	9	Austria

Question 3 (Easy)

What are the top three values the total invoices ?

```
---Q3: what are top 3 values of total invoice
select * from invoice
order by total desc
limit 3
```

Output:

total
double precision 
23.75999999999999998
19.8
19.8



Question 4 (Easy)

Which city has the best customers? we should like to throw a promotional music festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice total?

---Q4: Which city has the best customers? we should like to throw a promotional music festival in the city we made the most money
---Write a query that returns one city that has the highest sum of invoice totals.
---Return both the city name & sum of all invoice total

```
select sum (total) as invoice_total, billing_city
from invoice
group by billing_city
order by invoice_total desc
```

Output:

	invoice_total double precision	billing_city character varying (30)
1	273.24000000000007	Prague
2	169.29	Mountain View
3	166.32	London
4	158.4	Berlin
5	151.47	Paris
6	129.69	São Paulo
7	114.83999999999997	Dublin
8	111.86999999999999	Delhi
9	108.89999999999998	São José dos Campos
10	106.91999999999999	Brasília



Question 5 (Easy)

Who is the best customer? The customer who has spent the most money will be decleared the best customer.
Write a query that returns the person who has spent the most money?



```
---Q5:Who is the best coustomer? The customer who has spent the most money will be decleared the best customer.
----Write a query that returns the person who has spent the most money.
select customer.customer_id, customer.first_name,customer.last_name,sum(invoice.total) as total
from customer
join invoice on customer.customer_id=invoice.customer_id
group by customer.customer_id
order by total desc
limit 1|
```

Output:

	customer_id [PK] integer	first_name character (50)	last_name character (50)	total double precision
1	5	R...	Madhav...	144.540000000000002



Question 1 (MODERATE)

Write query to return the email, last name, first name,& Genre of all Rock Music Listeners. Return your list ordered alphabetically by email starting with A

```
--Q1: Write query to return the email, last name,first name,& Genre of all Rock Music Listeners.
-----Return your list ordered alphabetically by email starting with A
select distinct email, first_name, last_name
from customer
join invoice on customer.customer_id = invoice.customer_id
join invoice_line on invoice.invoice_id = invoice_line.invoice_id
where track_id in(
    select track_id from track
    join genre on track.genre_id = genre.genre_id
    where genre.name like 'Rock'
)
order by email;
```

Output:

	email character varying (50) 	first_name character (50) 	last_name character (50) 
1	aaronmitchell@yahoo.ca	Aaron ...	Mitchell ...
2	alero@uol.com.br	Alexandre ...	Rocha ...
3	astrid.gruber@apple.at	Astrid ...	Gruber ...
4	bjorn.hansen@yahoo.no	Bjørn ...	Hansen ...
5	camille.bernard@yahoo.fr	Camille ...	Bernard ...
6	daan_peeters@apple.be	Daan ...	Peeters ...
7	diego.gutierrez@yahoo.ar	Diego ...	Gutiérrez ...
8	dmiller@comcast.com	Dan ...	Miller ...
9	dominiquedefebvre@gmail.c...	Dominique ...	Lefebvre ...
10	edfrancis@yahoo.ca	Edward ...	Francis ...



Question 2 (MODERATE)

Let’s invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands?

```
---Q2: Lets invite the artists who have written the most rock music in our dataset.
---Write a query that returns the Artist name and total track count of the top 10 rock bands.
select artist.artist_id,artist.name,count(artist.artist_id) as number_of_songs
from track
join album on album.album_id = track.album_id
join artist on artist.artist_id = album.artist_id
join genre on genre.genre_id = track.genre_id
where genre.name like 'Rock'
group by artist.artist_id
order by number_of_songs desc
limit 10;
```

:Output



	artist_id [PK] character varying (50) 	name character varying (120) 	number_of_songs bigint 
1	22	Led Zeppelin	114
2	150	U2	112
3	58	Deep Purple	92
4	90	Iron Maiden	81
5	118	Pearl Jam	54
6	152	Van Halen	52
7	51	Queen	45
8	142	The Rolling Stones	41
9	76	Creedence Clearwater Revival	40
10	52	Kiss	35

Question 3 (MODERATE)

Return all the track names that have a song length longer than the average song length. Return the name and millisecond for each track. Order by the song length with the longest songs listed first.

select name , milliseconds?

```
---Q3: Return all the track names that have a song length longer than the average song length.
--Return the name and miliseconds for eack track .
--Order by the song length with the longest songs listed first.
select name,milliseconds
from track
where milliseconds>(
    select avg(milliseconds) as avg_track_length
    from track)
order by milliseconds desc;
```

Output:

	name character varying (150)	milliseconds integer
1	Occupation / Precipice	5286953
2	Through a Looking Glass	5088838
3	Greetings from Earth, Pt. 1	2960293
4	The Man With Nine Lives	2956998
5	Battlestar Galactica, Pt. 2	2956081
6	Battlestar Galactica, Pt. 1	2952702
7	Murder On the Rising Star	2935894
8	Battlestar Galactica, Pt. 3	2927802
9	Take the Celestra	2927677
10	Fire In Space	2926593



Question 1 (ADVANCE)

Find how much amount spent by each customer on artists? Write a query to return customer name, artist name and total spent?

```
---Q1:Find how much amount spent by each customer on artists?
---Write a query to return customer name, artist name and total spent?
WITH best_selling_artist AS (
    SELECT artist.artist_id AS artist_id, artist.name AS artist_name,
    SUM(invoice_line.unit_price*invoice_line.quantity) AS total_sales
    FROM invoice_line
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN album ON album.album_id = track.album_id
    JOIN artist ON artist.artist_id = album.artist_id
    GROUP BY 1
    ORDER BY 3 DESC
    LIMIT 1
)
SELECT c.customer_id, c.first_name, c.last_name, bsa.artist_name,
SUM(il.unit_price*il.quantity) AS amount_spent
FROM invoice i
JOIN customer c ON c.customer_id = i.customer_id
JOIN invoice_line il ON il.invoice_id = i.invoice_id
JOIN track t ON t.track_id = il.track_id
JOIN album alb ON alb.album_id = t.album_id
JOIN best_selling_artist bsa ON bsa.artist_id = alb.artist_id
GROUP BY 1,2,3,4
ORDER BY 5 DESC;
```



Question 1 (ADVANCE)

Find how much amount spent by each customer on artists?
Write a query to return customer name, artist name and total spent?

Output:

	customer_id 	first_name 	last_name 	artist_name 	amount_spent 
	integer	character (50)	character (50)	character varying (120)	double precision
1	46	Hugh ...	O'Reilly ...	Queen	27.719999999999985
2	38	Niklas ...	Schröder ...	Queen	18.81
3	3	François ...	Tremblay ...	Queen	17.82
4	34	João ...	Fernandes ...	Queen	16.830000000000002
5	53	Phil ...	Hughes ...	Queen	11.88
6	41	Marc ...	Dubois ...	Queen	11.88
7	47	Lucas ...	Mancini ...	Queen	10.89
8	33	Ellie ...	Sullivan ...	Queen	10.89
9	20	Dan ...	Miller ...	Queen	3.96
10	5	R ...	Madhav ...	Queen	3.96



Question 2 (ADVANCE)

We want to find out the most popular music Genre for each country , We determine the most popular genre as the genre With the highest amount of purchases , Write a query that returns each country along with the top Genre. For countries where ,The maximum number of purchases is shared return all Genres?

```

---Q2:We want to find out the most popular music Genre for each country?
---We determine the most popular genre as the genre ?
---With the highest amount of purchases?
---Write a query that returns each country along with the top Genre. For countries where?
---The maximum number of purchases is shared return all Genres?
WITH popular_genre AS
(
    SELECT COUNT(invoice_line.quantity) AS purchases, customer.country, genre.name, genre.genre_id,
    ROW_NUMBER() OVER(PARTITION BY customer.country ORDER BY COUNT(invoice_line.quantity) DESC) AS RowNo
    FROM invoice_line
    JOIN invoice ON invoice.invoice_id = invoice_line.invoice_id
    JOIN customer ON customer.customer_id = invoice.customer_id
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN genre ON genre.genre_id = track.genre_id
    GROUP BY 2,3,4
    ORDER BY 2 ASC, 1 DESC
)
SELECT * FROM popular_genre WHERE RowNo <= 1
    
```



Question 2 (ADVANCE)

We want to find out the most popular music Genre for each country , We determine the most popular genre as the genre With the highest amount of purchases , Write a query that returns each country along with the top Genre. For countries where ,The maximum number of purchases is shared return all Genres?

	purchases bigint	country character varying (50)	name character varying (120)	genre_id character varying (50)	rowno bigint
1	17	Argentina	Alternative & Punk	4	1
2	34	Australia	Rock	1	1
3	40	Austria	Rock	1	1
4	26	Belgium	Rock	1	1
5	205	Brazil	Rock	1	1
6	333	Canada	Rock	1	1
7	61	Chile	Rock	1	1
8	143	Czech Republic	Rock	1	1
9	24	Denmark	Rock	1	1
10	46	Finland	Rock	1	1



Question 3 (ADVANCE)

Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount?

```
--Q3:Write a query that determines the customer that has spent the most on music for each country.  
--Write a query that returns the country along with the top customer and how much they spent.  
--For countries where the top amount spent is shared, provide all customers who spent this amount.  
WITH Customer_with_country AS (  
    SELECT customer.customer_id,first_name,last_name,billing_country,SUM(total) AS total_spending,  
    ROW_NUMBER() OVER(PARTITION BY billing_country ORDER BY SUM(total) DESC) AS RowNo  
    FROM invoice  
    JOIN customer ON customer.customer_id = invoice.customer_id  
    GROUP BY 1,2,3,4  
    ORDER BY 4 ASC,5 DESC)  
SELECT * FROM Customer_with_country WHERE RowNo <= 1
```



Question 3 (ADVANCE)

Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount?

Output:

	customer_id integer	first_name character (50)	last_name character (50)	billing_country character varying (30)	total_spending double precision	rowno bigint
1	56	Diego	Gutiérrez	Argentina	39.6	1
2	55	Mark	Taylor	Australia	81.18	1
3	7	Astrid	Gruber	Austria	69.3	1
4	8	Daan	Peeters	Belgium	60.389999999999999	1
5	1	Luís	Gonçalves	Brazil	108.89999999999998	1
6	3	François	Tremblay	Canada	99.99	1
7	57	Luis	Rojas	Chile	97.020000000000001	1
8	5	R	Madhav	Czech Republic	144.540000000000002	1
9	9	Kara	Nielsen	Denmark	37.619999999999999	1
10	44	Terhi	Hämäläinen	Finland	79.2	1



Thank You For Watching

